24CSA111 - DATABASE MANAGEMENT SYSTEM (DBMS)

DBMS – Unit 1

- Introduction
- Data Independence
- The Three Levels of Architecture
- External Level
- Conceptual Level
- Internal Level
- Client/Server Architecture
- System Structure
- Instance and Schema
- Data Models
- Types of DBMS

Introduction

Data can be defined as known facts that can be recorded and that have implicit meaning. Data usually refers to raw data, or unprocessed data.

e.g. Each student's test score is one piece of data.

When data is processed, organized, structured or presented in a given context so as to make it useful, it is called information.

e.g. The average score of a class or of the entire school is information that can be derived from the given data.

Concept of Database

A database is a collection of related data that is organized so that it can be easily accessed, managed and updated.

Data is organized into rows, columns and tables, and it is indexed to make it easier to find relevant information.

Data gets updated, expanded and deleted as new information is added.

Database Management System

A database management system (DBMS) is a collection of programs that enables users to create and maintain a database.

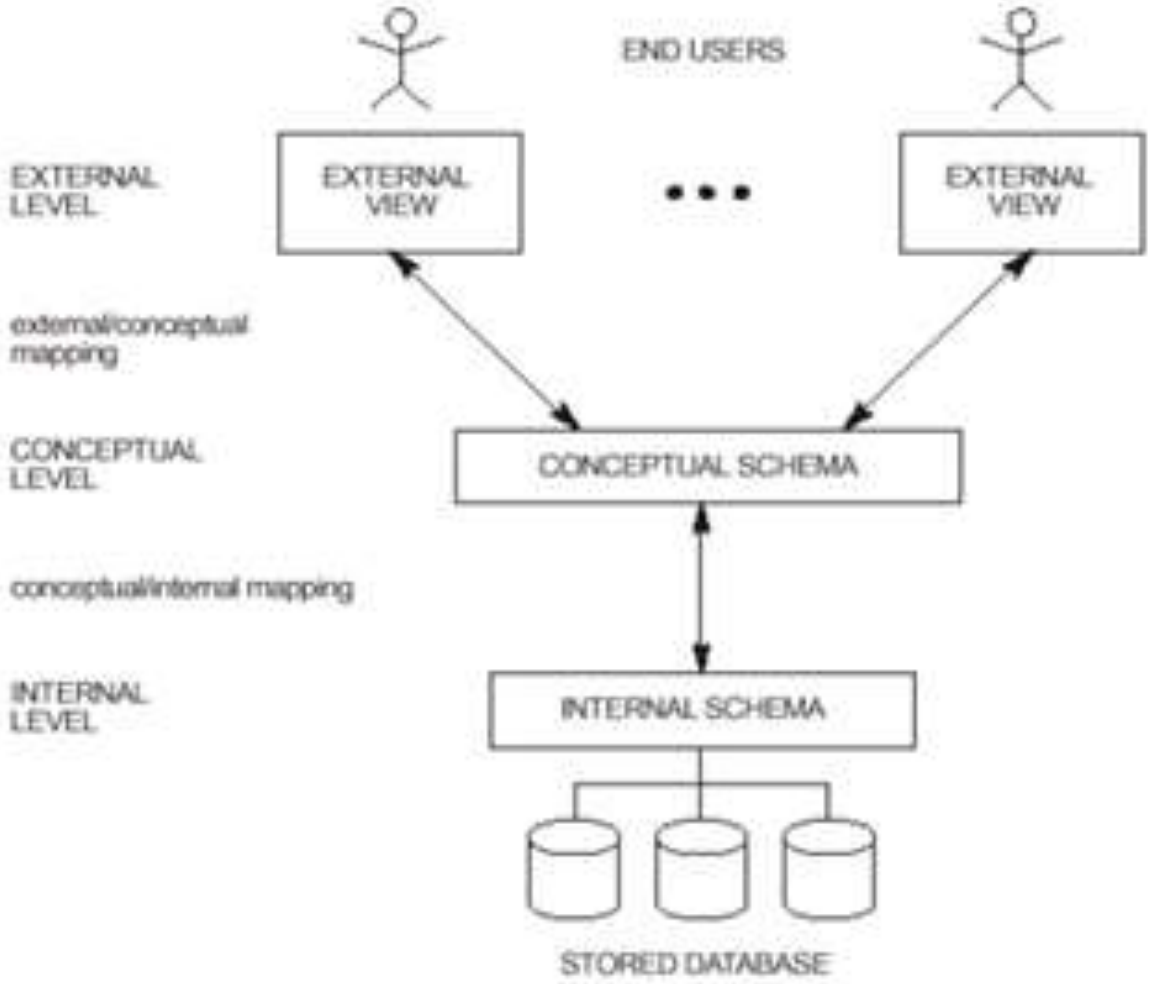
It is a general-purpose software system that facilitates the processes of defining, constructing, manipulating and sharing databases among various users and applications.

Three-Schema Architecture

The goal of the three-schema architecture is to separate the user applications from physical database. In this architecture, schemas can be defined at the following three levels:

- Internal level
- Conceptual level
- External level

Three-Schema Architecture



Three-Schema Architecture

- The Internal level has an internal schema, which describes the physical storage structure of the database.
- The conceptual level has a conceptual schema, which describes the structure of the whole database for a community of users.
- The external or view level includes a number of external schemas or user views.

Data Independence

It can be defined as the capacity to change the schema at one level of a database system without having to change the schema at the next higher level.

Two types of data independence:

- Logical data independence
- Physical data independence

Logical Data Independence

It is the capacity to change the conceptual schema without having to change external schemas or application programs.

(e.g. may expand the database by adding a record type or data item, changing the constraints or to reduce the database by removing a record type or data item)

Physical Data Independence

It is the capacity to change the internal schema without having to change the conceptual schema.

(e.g. creating additional access structures to improve the performance of retrieval or update of data)

Client/Server Architecture

This was developed to deal with computing environments in which a large number of PCs, workstations, file servers, printers, database servers and other software and equipment are connected via a network.

It is possible to connect a number of PCs or small workstations as clients to a file server that maintains the files of the client machines.

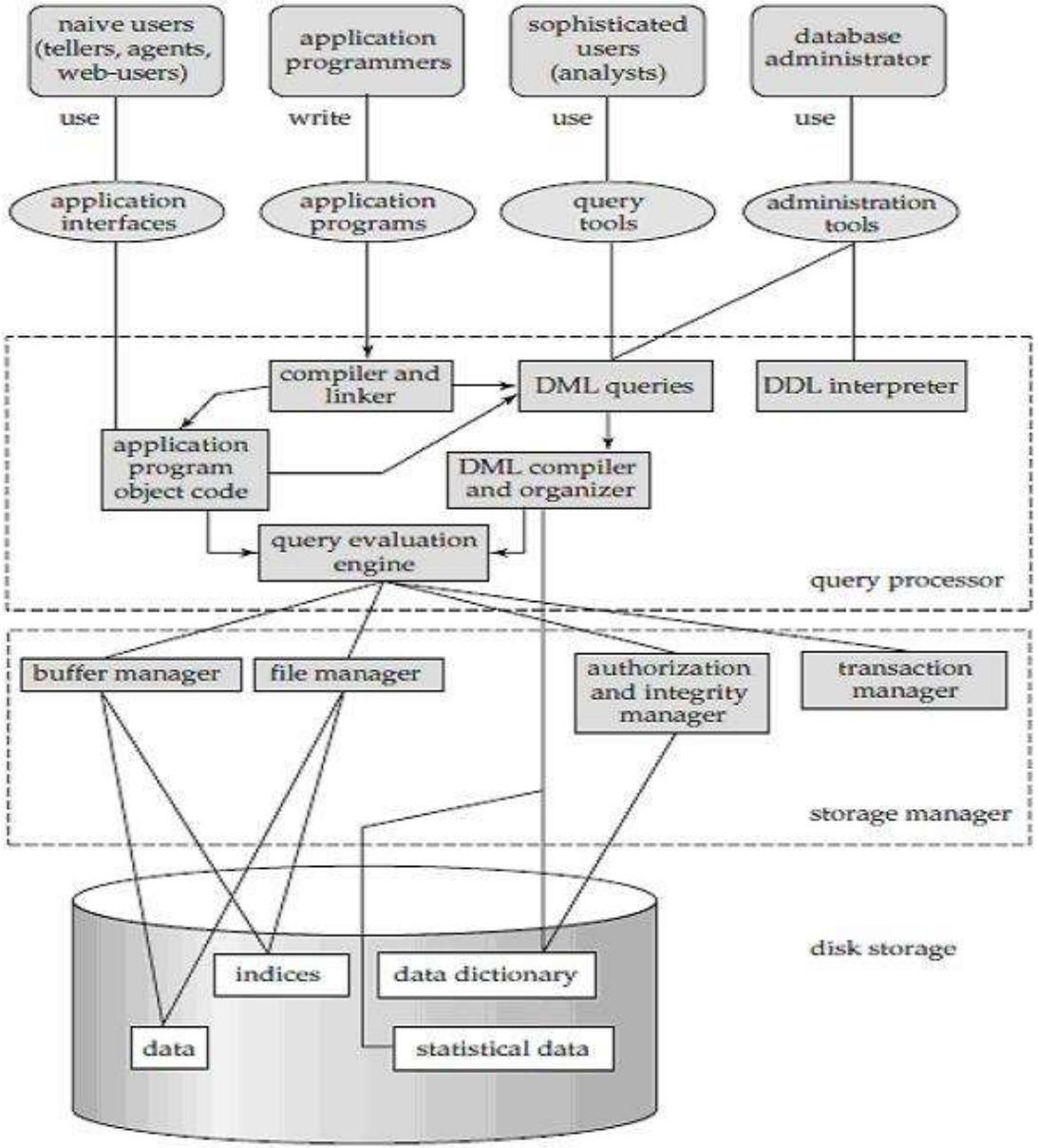
System Structure

A database system is partitioned into modules that deal with each of the responsibilities of the overall system.

The functional components of a database system can be broadly divided into the **storage manager** and the **query processor** components.

The storage manager is important because databases typically require a large amount of storage space. The query processor is important because it helps the database system simplify and facilitate access to data.

System Structure



System Structure Modules

Query Processor – includes compiler and linker, DML queries, DDL interpreter, DML compiler and organizer, application program object code and query evaluation engine.

Storage manager – includes buffer manager, file manager, authorization and integrity manager and transaction manager.

Data Dictionary (or metadata) – includes information such as the names and sizes of files, names and data types of data items, storage details of each file, mapping information among schemas and constraints.

Query Compiler – compiles the parsed query into an internal form and the same is optimized for a better performance.

Instance and Schema

A database instance is a set of memory structures that manage database files. The instance manages its associated data and serves the users of the database.

The overall design of a database is called schema. A schema contains schema objects like table, foreign key, primary key, views, columns, data types, stored procedures etc.

Data Models

A database model shows the logical structure of a database, including the relationships and constraints that determine how data can be stored and accessed.

Types of DBMS

Hierarchical databases

Network databases

Relational databases

Object-Oriented databases

Hierarchical databases

A DBMS is said to be hierarchical if the relationships among data in the database are established in such a way that one data item is present as the subordinate of another one.

Network databases

A DBMS is said to be a Network DBMS if the relationships among data in the database are of type of many-to-many.

Relational databases

A DBMS is said to be a Relational DBMS or RDBMS if the database relationships are treated in the form of a table.

Object-Oriented databases

Object-oriented databases use small, reusable chunks of software called objects. The objects themselves are stored in the object-oriented database.

THANK YOU